Hospital Healthcare Management: Findings Report

Oct 12, 2026

A review of 2,506 patient records from 2007 across 10 departments: who the patients are, where they were treated, how they fared, how they rated their care and what it cost.

Summary

The hospital treated 2,506 patients in 2007 at a total treatment cost of 404.5K (161 per patient on average) with a mean stay of 1.7 days. Eight findings frame everything that follows:

1. **Rehabilitation readmissions are a children's pattern.** 261 of the 265 children under 6 treated in Rehabilitation are readmitted, and they make up 53% of all 493 readmissions in the hospital.
2. **Deaths rise sharply with age.** There are no deaths under 41, 6.3% of patients aged 41 to 60 and 22.3% of those over 60 die, and every death is in Rehabilitation (52) or Operation Theatre (34).
3. **March was two bursts, not a steady month.** 93% of March's 904 patients arrived in 15 days (1 to 10 and 21 to 25 March), led by OPD and the smaller specialties.
4. **Surgery and Rehabilitation use 54% of bed-days** (1,230 and 1,077 of 4,253), while OPD, with a third of the patients, uses 2%.
5. **Cost follows volume.** Average cost per patient is almost identical across the four busiest departments (153 to 158) and across patient types (165 inpatient, 154 outpatient).
6. **Cost is concentrated.** 190 treatments above 500 are 8% of patients but 38% of total cost.
7. **Three departments dominate.** OPD, Rehabilitation and Surgery see 71% of patients and generate 67% of cost.
8. **Satisfaction is high.** 96% of feedback is positive and the average rating is 4.75 out of 5; Oncology (60% positive) and Neurology (75%) are lowest.

Overall picture

Activity is dominated by a single month: March accounts for 36% of the year's patients and cost, and every other month stays between 88 and 285 patients.

Month (2007)	Patients	Cost (K)	Avg cost per patient	Inpatient share
January	148	27.0	182	67%
February	123	17.5	143	92%
March	904	147.3	163	57%
April	141	22.9	162	74%
May	200	30.5	153	74%
June	285	50.0	175	66%
July	132	19.2	145	72%
August	201	32.6	162	87%
September	97	14.6	150	77%
October	88	11.8	135	74%
November	99	18.3	185	68%
December	88	12.7	145	36%

- **Volume sets monthly cost.** Average cost per patient moves in a narrow band (135 to 185), so months with more patients cost more almost one for one.
- **Peaks and troughs.** After March, June is the next peak (285 patients, 50K); October and December are the quietest (88 patients each).
- **Patient mix.** 1,679 patients (67%) are inpatients and 827 (33%) are outpatients. December has the lowest inpatient share (36%), February the highest (92%).
- **Length of stay.** 755 patients (30%) stay 0 days, 1,090 (43%) stay 2 days and 267 inpatients stay 3 days or more, up to a maximum of 32.
- **Outcome mix.** 65.3% have a normal outcome, 19.7% are readmitted, 8.3% are discharged, 3.4% die and 3.3% go to ICU.

Patients

Patients are young (average age 30.5, oldest 98) and almost evenly split by gender, and the care setting depends strongly on age.

Age group	Patients	Share	Inpatients	Outpatients	Total cost (K)	Avg cost per patient
0 to 6	317	12.7%	291	26	51.9	164
6 to 20	794	31.7%	480	314	136.0	171
21 to 40	512	20.4%	257	255	76.9	150
41 to 60	695	27.7%	464	231	110.3	159
60+	188	7.5%	187	1	29.4	157

- **Age.** The 6 to 20 group is the largest (794 patients, 32%) and also the costliest group (136K, 34% of total cost).
- **Setting by age.** 92% of children under 6 (291 of 317) and 99.5% of patients over 60 (187 of 188) are inpatients, against 60% for ages 6 to 20 and 50% for ages 21 to 40.
- **Gender.** 1,280 patients are female (51.1%) and 1,226 male (48.9%). Cost is nearly identical: 205.9K for women and 198.5K for men, about 161 to 162 per patient.
- **Geography.** Patients come from 37 states and regions. British Columbia alone supplies 977 (39%), followed by California (402), Washington (215), England (175) and New South Wales (130); these five account for 1,899 patients (76%).
- **Rating by setting.** Outpatients rate their care slightly higher (4.81) than inpatients (4.72).

Departments

Three departments (OPD, Rehabilitation and Surgery) see 71% of patients and generate 67% of cost, while the smallest departments cost the most per patient.

Department	Patients	Total cost (K)	Share of cost	Avg cost per patient	Median cost	Avg stay (days)
OPD	824	126.2	31.2%	153	44.7	0.1
Rehabilitation	515	79.5	19.7%	154	44.7	2.1
Surgery	436	66.9	16.5%	153	43.7	2.8
Operation Theatre	242	38.1	9.4%	158	44.7	2.0

Department	Patients	Total cost (K)	Share of cost	Avg cost per patient	Median cost	Avg stay (days)
Dermatology	169	31.8	7.9%	188	44.7	2.9
Orthopaedics	96	15.8	3.9%	165	47.4	2.2
Neurology	72	11.6	2.9%	161	62.1	3.8
Gynaecology	59	12.7	3.1%	215	152.4	3.9
Cardiology	48	12.8	3.2%	266	50.6	2.0
Oncology	45	9.2	2.3%	205	73.6	2.0
All departments	**2,506**	**404.5**	**100%**	**161**	**44.7**	**1.7**

- **Concentration.** OPD, Rehabilitation and Surgery together treat 1,775 patients and account for 272.5K of cost.
- **Cost per patient rises as volume falls.** Cardiology (266), Gynaecology (215) and Oncology (205) are the costliest per patient but see 59 patients or fewer.
- **What drives the averages.** Cardiology's median is only 50.6 against a mean of 266, so a few very expensive cases lift it. Gynaecology is expensive throughout, with a median of 152, more than three times that of most departments.
- **Length of stay.** OPD averages 0.1 days. Gynaecology (3.9) and Neurology (3.8) have the longest stays, followed by Dermatology (2.9) and Surgery (2.8).
- **Staff.** All 262 staff members treated patients in each of OPD, Rehabilitation and Surgery, so staff work across departments rather than belonging to one.

Bed-days by department

Bed-days (the sum of length of stay) show where hospital capacity is used: two departments take more than half of it.

Department	Bed-days	Share of bed-days	Patients	Avg stay (days)
Surgery	1,230	28.9%	436	2.8
Rehabilitation	1,077	25.3%	515	2.1
Dermatology	485	11.4%	169	2.9
Operation Theatre	477	11.2%	242	2.0
Neurology	274	6.4%	72	3.8

Department	Bed-days	Share of bed-days	Patients	Avg stay (days)
Gynaecology	231	5.4%	59	3.9
Orthopaedics	212	5.0%	96	2.2
Cardiology	96	2.3%	48	2.0
Oncology	89	2.1%	45	2.0
OPD	82	1.9%	824	0.1
All departments	**4,253**	**100%**	**2,506**	**1.7**

- **Surgery and Rehabilitation** use 54% of all bed-days (2,307 of 4,253) while treating 38% of patients.
- **OPD** treats a third of all patients (824) but uses under 2% of bed-days.
- **Long-stay specialties.** Neurology and Gynaecology are 5% of patients but 12% of bed-days, and Dermatology is 7% of patients but 11% of bed-days.
- **Inpatients** account for 4,171 of the 4,253 bed-days (98%).

Outcomes and Experience

Every readmission and death sits in a few departments, and satisfaction is high everywhere except Oncology and Neurology.

Clinical outcomes

Department	Normal	Readmit	Discharge	ICU	Death
OPD	756	66	1	1	0
Rehabilitation	70	393	0	0	52
Surgery	182	22	190	42	0
Operation Theatre	196	12	0	0	34
Dermatology	160	0	0	9	0
Orthopaedics	75	0	18	3	0
Neurology	66	0	0	6	0
Gynaecology	51	0	0	8	0
Cardiology	40	0	0	8	0

Department	Normal	Readmit	Discharge	ICU	Death
Oncology	40	0	0	5	0
All departments	**1,636**	**493**	**209**	**82**	**86**

- **Readmissions.** 493 patients (19.7%) were readmitted: 393 in Rehabilitation (a 76% readmission rate for that department), 66 in OPD (8%), 22 in Surgery (5%) and 12 in Operation Theatre (5%). No other department has any.
- **Deaths.** 86 patients died (3.4%): 52 in Rehabilitation (10% of its patients) and 34 in Operation Theatre (14%). All were aged 41 or older (44 aged 41 to 60 and 42 aged over 60).
- **ICU.** 82 patients (3.3%) went to ICU, 42 of them from Surgery; every department except Rehabilitation and Operation Theatre has some.
- **Discharges.** 190 of the 209 discharges (91%) are in Surgery, 18 in Orthopaedics and 1 in OPD.
- **Stay by outcome.** ICU patients stay longest (4.1 days), then discharged (3.4), deaths (2.0), readmissions (1.8) and normal outcomes (1.3).

Death rate by age

Mortality is a pattern of older patients: no one under 41 died, and the rate more than triples from the 41 to 60 group to the over-60 group.

Age group	Patients	Deaths	Death rate	ICU admissions	ICU rate
0 to 6	317	0	0%	5	1.6%
6 to 20	794	0	0%	40	5.0%
21 to 40	512	0	0%	33	6.4%
41 to 60	695	44	6.3%	1	0.1%
60+	188	42	22.3%	3	1.6%
All patients	**2,506**	**86**	**3.4%**	**82**	**3.3%**

- **Where the deaths are.** All 86 are in two departments. In Rehabilitation, 26 of 95 patients aged 41 to 60 (27%) and 26 of 78 aged over 60 (33%) died; in Operation Theatre, 18 of 124 (15%) and 16 of 106 (15%).
- **No deaths elsewhere.** The other 480 patients aged 41 or over, treated in the other eight departments, had none.
- **ICU is a younger outcome.** 78 of the 82 ICU patients are under 41; only 4 are older.

Patient satisfaction

Department	Positive feedback	Avg rating (of 5)
Surgery	100%	4.98
Gynaecology	100%	4.97
Orthopaedics	100%	4.94
Dermatology	97.6%	4.89
OPD	96.8%	4.81
Rehabilitation	92.2%	4.66
Cardiology	91.7%	4.60
Operation Theatre	100%	4.50
Neurology	75.0%	4.13
Oncology	60.0%	3.73

- **Overall.** 2,396 of 2,506 responses (95.6%) are positive: 1,828 Fully Agree (73%) and 568 Rather Agree (23%). 81 are negative (3.2%) and 29 are Don't know (1.2%). The average rating is 4.75 out of 5.
- **Lowest satisfaction.** Oncology has 18 Rather Disagree responses out of 45 (40%) and Neurology 18 negative out of 72 (25%).
- **Operation Theatre.** All 242 responses are Rather Agree, with none at Fully Agree, which matches its lower rating of 4.50.
- **Strongest disagreement.** Of the 30 Fully Disagree responses, 21 are in OPD, 4 in Cardiology, 3 in Dermatology and 2 in Neurology.
- **Don't know.** All 29 come from Rehabilitation.

Rehabilitation deep-dive

Rehabilitation treats 515 patients (21% of the hospital) and is where outcomes diverge most: 76% are readmitted, 10% die and only 14% have a normal outcome. The pattern depends almost entirely on age.

Age group	Patients	Normal	Readmit	Death
0 to 6	265	4	261	0
6 to 20	51	0	51	0
21 to 40	26	0	26	0
41 to 60	95	32	37	26
60+	78	34	18	26
All ages	**515**	**70**	**393**	**52**

- **Children drive readmissions.** 265 of the 515 patients (51%) are under 6, and 261 of them are readmitted (98.5%). That is 53% of all readmissions in the hospital and 82% of all children under 6 treated anywhere.
- **Under 41 versus over 40.** Of 342 patients under 41, 338 (99%) are readmitted. Of 173 aged over 40, 66 (38%) have a normal outcome, 55 (32%) are readmitted and 52 (30%) die.
- **Timing.** August has 113 readmissions among 118 patients (96%); May and June have 62 each. These three months hold 237 of the 393 readmissions (60%).
- **Length of stay.** All Rehabilitation patients are inpatients, and 485 of 515 (94%) stay exactly 2 days. Readmitted patients average 2.0 days, normal outcomes 2.7 and deaths 2.0.
- **Cost.** Rehabilitation readmissions cost 66.7K, 83% of all readmission cost (80.2K); children under 6 alone account for 43.7K (55%). A readmitted patient averages 169.7, against 113.5 for a normal outcome and 93.4 for a death.
- **Satisfaction.** 92.2% of feedback is positive, with an average rating of 4.66; all 29 "Don't know" answers in the hospital come from this department.
- **How to read this.** Nearly every child is readmitted, in most months, for a standard 2-day stay. That looks like recurring treatment cycles rather than failed discharges, so what "Readmit" means in the source should be confirmed before it is treated as a quality problem.

What drove March

March's 904 patients came in two bursts, not as a steady month: 837 of them (93%) arrived on 15 days, and the surge came from OPD and the smaller specialties, not from Rehabilitation or Operation Theatre.

Department	March patients	Average in other months	Times higher	Share of March	Share of rest of year
OPD	386	39.8	9.7x	42.7%	27.3%
Surgery	111	29.5	3.8x	12.3%	20.3%
Dermatology	104	5.9	17.6x	11.5%	4.1%
Orthopaedics	80	1.5	55x	8.8%	1.0%
Rehabilitation	62	41.2	1.5x	6.9%	28.3%
Neurology	51	1.9	26.7x	5.6%	1.3%
Gynaecology	39	1.8	21.4x	4.3%	1.2%
Oncology	27	1.6	16.5x	3.0%	1.1%
Cardiology	26	2.0	13.0x	2.9%	1.4%
Operation Theatre	18	20.4	0.9x	2.0%	14.0%
All departments	**904**	**145.6**	**6.2x**	**100%**	**100%**

- **Two windows.** 1 to 10 March has 515 patients (about 50 a day, peaking at 77 on the 10th) and 21 to 25 March has 322 (58 to 69 a day). The other 16 days average 4 patients.
- **Size of the surge.** March had about 758 more patients than a typical month (904 against 146).
- **Who came.** OPD is 43% of March patients against 27% in the rest of the year, and six smaller departments each ran 13 to 55 times their usual monthly count.
- **Age and setting.** Patients aged 41 to 60 are 41% of March (20% otherwise) and children under 6 only 5% (17% otherwise); 43% are outpatients against 28% in the rest of the year.
- **Outcomes and cost.** 82% of March patients had a normal outcome against 56% in the rest of the year, with only 30 readmissions. March cost 147.3K, of which OPD 58.8K (40%), Dermatology 19.0K and Surgery 18.0K.
- **How to read this.** Two short windows and a jump in departments that are otherwise small is consistent with a screening or outreach campaign, or with a batch of records entered together; the data alone cannot say which, so the cause should be confirmed

with the source.

Treatment Costs

Total cost is 404.5K, but a typical treatment is small and a few expensive ones carry the total: the median is 44.7 against a mean of 161, and the highest single treatment is 1,241.

Patient group	Patients	Total cost (K)	Share of cost	Avg cost per patient
Inpatients	1,679	277.4	68.6%	165
Outpatients	827	127.1	31.4%	154
Normal outcome	1,636	264.0	65.3%	161
Readmit	493	80.2	19.8%	163
Discharge	209	31.1	7.7%	149
ICU	82	19.9	4.9%	242
Death	86	9.3	2.3%	108

- **Concentration.** 190 treatments above 500 (8% of patients) account for 154.2K, 38% of all cost.
- **Inpatients and outpatients.** Inpatients carry 69% of cost simply because they are 67% of patients; the average per patient differs by only 11 (165 against 154).
- **ICU.** ICU is the most expensive outcome per patient (242), 50% above a normal case (161).
- **Readmissions.** They cost 80.2K, one fifth of all spend, and most of it comes from Rehabilitation, which holds 393 of the 493 readmissions.
- **Deaths.** These patients cost the least on average (108), consistent with their short stays (2.0 days).
- **Age and gender.** The 6 to 20 group costs the most (136K, 34%), followed by 41 to 60 (110K, 27%); women and men cost almost the same (205.9K and 198.5K).
- **Month.** March costs 147K (36% of the year); the next highest month, June, costs 50K.

Cost of readmissions (estimate)

Costs in this dataset carry no stated currency, so all cost figures in this report are in the source's own units. The figures below are an illustrative estimate, not a measured result.

- **Basis.** Rehabilitation has 393 readmissions at an average treatment cost of 169.7, about 66.7K in total.
- **Scenario.** If readmissions fell by 25%, about 98 fewer would occur, saving about 16.7K (4.1% of the 404.4K total) and about 195 bed-days.
- **Assumption.** The estimate assumes each avoided readmission would not be replaced by another admission and that the average cost stays the same.

Notes and limitations

- **Single year.** The data covers 2007 only, so seasonal patterns such as March cannot be compared with other years.
- **Currency.** The source gives no currency for treatment cost.
- **Missing values.** 755 patients have no bed recorded and 719 have no ER time, so bed-day and ER figures cover the remaining patients only.
- **Outpatient stays.** OPD stays average 0.1 days, so bed-day figures are driven by inpatients.
- **Feedback.** Every Operation Theatre response is "Rather Agree", and all "Don't know" answers come from Rehabilitation, so those results may reflect how feedback was collected.
- **Readmit.** The source does not define "Readmit", so the Rehabilitation readmission rate depends on how the status was assigned.
- **March.** The data shows when the surge occurred but not why; the cause is unconfirmed.

Recommendations

The findings point to seven areas for action, in priority order.

1. **Investigate the Rehabilitation readmission cycle.** Readmissions are concentrated among children and recur; first confirm what "Readmit" means in the source, then review follow-up care. A 25% reduction would save about 16.7K and 195 bed-days (estimate).

2. **Review outcomes for older patients and Operation Theatre.** Death rates rise sharply with age, and Operation Theatre has the highest rate of any department (34 of 242, 14%).
3. **Plan capacity for March-type surges.** March had two bursts (1 to 10 and 21 to 25 March) with 36% of the year's patients and cost; confirm the cause and plan staff and beds around it.
4. **Plan beds around Surgery and Rehabilitation.** Together they use 54% of bed-days; Gynaecology (3.9 days) and Neurology (3.8) have the longest average stays.
5. **Follow up Oncology and Neurology feedback.** Positive rates of 60% and 75% are well below the 92% to 100% elsewhere; both departments are small, so collect more responses.
6. **Examine high-cost cases.** 190 treatments above 500 drive 38% of cost.
7. **Improve data capture.** Record currency, a readmission definition, and bed and ER time for every patient.

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